Power BI Dashboard Specification

Navigation Product Analytics — 4-Page Dashboard

Audience: Product leadership, Product Analytics, Growth

Refresh cadence: Daily

Data sources: navigation_sessions, route_options, user_events, experiments, retention_daily, users

Page 1 — Executive Product Overview

Business Questions Answered

- How is the business trending overall — users, sessions, and core quality metrics?
- What is the current recommended-route acceptance rate and how has it moved over time?
- How healthy is navigation completion and retention at a glance?

KPIs (Cards)

- Total Active Users (period)
- Total Navigation Sessions
- Recommended-Route Acceptance Rate
- Navigation Completion Rate
- 7-Day Retention Rate
- 30-Day Retention Rate

Charts & Visuals

- Line chart: Daily sessions and daily acceptance rate over time (dual axis)
- KPI trend sparklines for each headline metric vs. prior period
- Map or bar chart: sessions by city
- Donut chart: session outcome mix (Completed / Abandoned / Not Started)

Filters / Slicers

- Date range
- City
- Country
- Device type
- Subscription type
- User type (New/Returning)

Key Dimensions

date, city, country, device_type, subscription_type, user_type

Key Measures (DAX concepts, not formulas)

- Acceptance Rate = Accepted Sessions / Total Sessions
- Completion Rate = Completed Navigations / Started Navigations
- Active Users = DISTINCTCOUNT(user_id) over selected period

- Retention Rate = Retained Users / Cohort Size (from retention_daily)

Page 2 — Route Recommendation Analytics

Business Questions Answered

- Does toll presence influence route acceptance?
- Does traffic level or ETA error influence rejection?
- Which route types are recommended most, and which are actually selected most?
- How accurate are ETAs, and where does accuracy break down?

KPIs (Cards)

- Acceptance Rate by Route Type
- Average ETA Error (minutes)
- Average Reroute Count
- Toll-Route Acceptance Rate vs. Toll-Free Acceptance Rate

Charts & Visuals

- Bar chart: Acceptance rate by recommended_route_type
- Bar chart: Acceptance rate by traffic_level and toll_recommended (small multiples)
- Scatter plot: traffic_uncertainty_score vs. acceptance rate (binned)
- Histogram: distribution of ETA error (actual_duration_minutes – recommended_eta_minutes)
- Heatmap: ETA error by city x traffic_level

Filters / Slicers

- Route type
- Traffic level
- Weather condition
- Toll recommended (Y/N)
- City
- Date range

Key Dimensions

route_type, traffic_level, weather_condition, toll_recommended, city

Key Measures (DAX concepts, not formulas)

- ETA Error = actual_duration_minutes – recommended_eta_minutes
- ETA Accuracy % = 1 – (ABS(ETA Error) / actual_duration_minutes)

- Toll Rejection Rate = Rejected Sessions WHERE toll_recommended = TRUE / Sessions WHERE toll_recommended = TRUE
- Avg Reroutes per Started Navigation

Page 3 — User Behavior & Retention

Business Questions Answered

- Do new and returning users behave differently?
- Which segments navigate most frequently and complete the most trips?
- How do DAU / WAU / MAU trend, and how does retention look by signup cohort?

KPIs (Cards)

- DAU (latest day)
- WAU (rolling 7-day)
- MAU (rolling 30-day)
- Avg Sessions per Active User
- 30-Day Retention Rate (latest complete cohort)

Charts & Visuals

- Line chart: DAU / WAU / MAU over time
- Cohort retention heatmap: signup month x days-since-signup (7 / 14 / 30 day markers)
- Bar chart: sessions per user by subscription_type and user_type
- Bar chart: acceptance rate and completion rate by preferred_route_type

Filters / Slicers

- Signup cohort (month)
- Subscription type
- User type
- City
- Preferred route type

Key Dimensions

signup_month, subscription_type, user_type, city, preferred_route_type

Key Measures (DAX concepts, not formulas)

- DAU = DISTINCTCOUNT(user_id) WHERE active_flag = 1, per day
- WAU / MAU = rolling distinct active users over a 7- or 30-day window
- Retention Day N = Active users N days after cohort reference date / cohort size
- Sessions per User = Total Sessions / Distinct Active Users

Page 4 — Experiment & Product Insights

Business Questions Answered

- Did the ETA-confidence-range treatment improve route acceptance?
- Were there any regressions on guardrail metrics (abandonment, completion)?
- What is the estimated retention impact of the treatment?

KPIs (Cards)

- Route Acceptance Rate — Control vs Treatment
- Navigation Completion Rate — Control vs Treatment
- 7-Day Retention — Control vs Treatment
- Statistical Significance Indicator (lift + confidence)

Charts & Visuals

- Bar chart: primary and secondary metrics side-by-side by experiment_group
- Line chart: cumulative acceptance rate by group over the experiment period (sanity/SRM check)
- Table: metric, Control value, Treatment value, absolute lift, relative lift
- Funnel visual: Control vs Treatment side-by-side funnel from exposure to retained_7d

Filters / Slicers

- Experiment group
- Date range (exposure_timestamp)
- City

Key Dimensions

experiment_group, exposure_date, city

Key Measures (DAX concepts, not formulas)

- Group Acceptance Rate = AVG(route_acceptance) BY experiment_group
- Lift = (Treatment Rate – Control Rate) / Control Rate
- Guardrail Flag = IF(Treatment abandonment rate significantly > Control, 'Regression', 'OK')